Comparative Study and Implementation of Generative AI Models

Assignment No. 2: AI-Assisted Comparative Study and Implementation of Generative AI Models | CO3 and CO4

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Abstract

This report compares four generative models for images: an Autoencoder (AE), a Variational Autoencoder (VAE), a Deep Convolutional GAN (DCGAN) and a pretrained Stable Diffusion v1.5 model. The AE, VAE and DCGAN were implemented in PyTorch and trained for 10 epochs on the Fashion-MNIST dataset in Google Colab on a Tesla T4 GPU. Stable Diffusion was used for inference only, with four product-photography prompts. The report explains how each model works, compares their architectures, strengths and limitations, and analyses the reconstructed and generated outputs. On the 10,000-image test set the AE reached a reconstruction MSE of 0.009628 and the VAE 0.015939, and the visual blur seen in the VAE outputs agrees with this gap. The DCGAN produced recognisable clothing shapes but many noisy samples, while Stable Diffusion produced the most realistic images, with some prompt mismatches. The study is limited by short training, a single run and different data across the models.

1. Introduction

Generative AI models learn the structure of data so that they can rebuild or create content. In image work, two goals need to be kept apart. Reconstruction means that a model receives an image, compresses it into a small code and rebuilds it, so the output can be compared pixel by pixel with the input. Generation means that a model creates a new image from random noise or from a text prompt, so there is no single correct output to compare against. Autoencoders and variational autoencoders are mainly studied for reconstruction, while GANs and diffusion models are mainly studied for generation, although a VAE can also generate by sampling.

This matters in practice because the right way to judge a model depends on its goal. A reconstruction error such as mean squared error (MSE) is meaningful for an AE, but it says little about how realistic a GAN sample is. For text-to-image models, realism and agreement with the prompt matter more than any pixel error. This report therefore compares the four models on their principles, architectures and observed outputs, and is careful not to rank models on a number that does not apply to all of them.

1.1 Aim

To compare the working principles, architectures, strengths and limitations of an Autoencoder, a Variational Autoencoder, a DCGAN and a pretrained Stable Diffusion model, and to analyse their reconstructed and generated outputs from a practical Google Colab experiment.

1.2 Objectives

- Run AE and VAE reconstruction on the same test images and compare blur, distortion and loss of detail (CO3).
- Train a compact DCGAN and inspect the quality of its generated samples (CO4).
- Generate images with the pretrained Stable Diffusion v1.5 model from text prompts (CO4).
- Compare all outputs and state the limits of the comparison honestly, without overstating the results.

2. Background and Working Principles

2.1 Autoencoder (AE)

An autoencoder has two parts. The encoder maps an input image x to a short latent vector z, and the decoder maps z back to a reconstruction $\hat{x}$. Because z is much smaller than the input, the network is forced to keep only the most useful information [1]. In our notebook, a 28 × 28 image (784 values) is compressed to 64 values and then expanded again. Training minimises the average squared difference between input and output:

$$L_AE = (1/N)\ \Sigma\ \| x - \hat{x} \|^2$$

Since the loss only rewards copying the input, an AE has no reason to arrange its latent space in an orderly way. Nearby latent points do not have to decode to similar images, so picking a random latent vector usually gives a poor image. This is why a plain AE is used for compression, denoising and anomaly detection rather than for generating new samples.

2.2 Variational Autoencoder (VAE)

A VAE changes the encoder so that it outputs a distribution rather than a single point: a mean μ and a log-variance $\log \sigma^2$ for each latent value [2]. A latent vector is then sampled using the reparameterisation trick, which keeps the sampling step differentiable:

$$z = \mu + \sigma \odot \varepsilon, \quad \varepsilon \sim N(0, I)$$

The loss has two terms. The first is a reconstruction term, which in our notebook is binary cross-entropy summed over all pixels. The second is the Kullback–Leibler (KL) divergence, which pulls each latent distribution towards a standard normal distribution:

$$L_VAE = BCE(x, \hat{x}) + KL = BCE(x, \hat{x}) - \tfrac{1}{2}\ \Sigma\ (1 + \log \sigma^2 - \mu^2 - \sigma^2)$$

The KL term gives the latent space a smooth, regular shape, so that points drawn from a normal distribution decode to plausible images. The price is that the sampled latent vector adds noise during reconstruction and the KL pressure limits how much detail can be stored, which usually makes VAE images blurrier than AE images.

2.3 Deep Convolutional GAN (DCGAN)

A generative adversarial network trains two networks against each other [3]. The generator G turns a random noise vector into an image. The discriminator D receives real and generated images and tries to tell them apart. D is trained to label real images as real and generated images as fake, while G is trained so that D labels its images as real. The original objective is a minimax game:

$$min_G\ max_D\ \ E[\ \log D(x)\ ] + E[\ \log (\ 1 - D(\ G(z)\ )\ )\ ]$$

A DCGAN applies this idea with convolutional layers: transposed convolutions in the generator to grow an image, strided convolutions in the discriminator to shrink it, batch normalisation in both, and a Tanh output so that pixels lie in [−1, 1] [4]. No input image is compared with the output, so there is no reconstruction error. The two losses depend on each other, which makes GAN training delicate. Known problems include unstable training, artefacts and mode collapse, where the generator produces only a few kinds of output.

2.4 Pretrained Stable Diffusion

Diffusion models learn to reverse a gradual noising process [5]. During training, noise is added to images step by step, and a network learns to predict and remove that noise. To generate an image, the model starts from pure noise and removes it step by step. Stable Diffusion is a latent diffusion model [6]: the denoising happens in a compressed latent space rather than in pixel space, which makes it much cheaper. A text encoder turns the prompt into embeddings that guide a U-Net denoiser through cross-attention, and the final latent is decoded to an image by the decoder of an autoencoder. The guidance scale controls how strongly the result follows the prompt. In this study we did not train this model; we loaded the pretrained v1.5 pipeline from Hugging Face [8] and ran it for inference.

3. Comparative Table

Aspect	Autoencoder (AE)	Variational Autoencoder (VAE)	DCGAN	Stable Diffusion (pretrained)
Primary purpose and output	Image reconstruction. Output: a reconstructed copy of the input image.	Reconstruction and generation by sampling the latent space. Output: reconstructed or new images.	Image generation from random noise. Output: new synthetic images.	Text-to-image generation. Output: an image matching the text prompt.
Working principle	Encoder compresses the image into a latent vector; decoder rebuilds it. Training minimises reconstruction error [1].	Encoder predicts mean and log-variance; z is sampled with the reparameterisation trick. Loss = reconstruction + KL divergence [2].	Generator maps noise to images while a discriminator separates real from fake; trained adversarially [3], [4].	Latent diffusion model removes noise step by step from a latent, guided by the encoded text prompt, then decodes it [5], [6].
Architecture (as in the notebook)	Fully connected: 784→256→64 (latent), then 64→256→784; ReLU, Sigmoid output.	Fully connected: 784→400, two heads for mean and log-variance (20 values each), then 20→400→784; Sigmoid output.	Generator: Linear(100→6272), 2 transposed-conv layers (BatchNorm, ReLU, Tanh). Discriminator: 2 conv layers + linear (LeakyReLU, Sigmoid).	Pretrained pipeline: text encoder, U-Net denoiser and autoencoder decoder, loaded from Hugging Face [8].
Training or inference here	10 epochs on 60,000 Fashion-MNIST images; Adam, lr 1e-3; MSE loss.	10 epochs on the same data; Adam, lr 1e-3; BCE + KL loss.	10 epochs on the same data; Adam, lr 2e-4; noise size 100; BCE adversarial loss.	No training. Inference only: 4 prompts, 20 steps, guidance scale 7.0, 512 × 512.
Advantages	Simple and fast to train; compact representation.	Smooth probabilistic latent space; can generate new samples.	Can give sharp samples; quick to sample once trained.	Detailed, realistic, text-controllable images without our own training.
Limitations	Blurring and loss of fine detail; latent space not designed for sampling.	Blurrier output than AE; random samples can be unclear.	Unstable training, artefacts, mode collapse; short training gives noisy samples.	Heavy compute (GPU); sensitive to prompt wording; licence rules; artefacts possible.
Applications	Compression, denoising, anomaly detection.	Representation learning, data synthesis, latent interpolation.	Data augmentation, image and texture synthesis.	Product mock-ups, concept art, design drafts, creative content.
Observed outputs (this study)	Shapes kept; fine texture softened (logo smeared, plaid faded); sandal and sneaker faint but recognisable.	Same shapes but visibly blurrier; logo and pattern lost; patterned shirt and sandal become smooth blobs.	Recognisable shoes, tops and trousers with fairly sharp edges; many noisy or broken samples.	Photorealistic colour product images with fine detail; some prompt mismatches and odd shapes.

Table 1 Comparison of the four models. Architecture and training details are taken from the notebook code and printed model summaries.

4. Implementation and Experimental Setup

4.1 Environment and data

All experiments ran in one Google Colab notebook using PyTorch 2.11.0 (CUDA 13.0 build) on an NVIDIA Tesla T4 GPU. Fashion-MNIST was downloaded through torchvision. It contains 60,000 training and 10,000 test images of size 28 × 28 in grayscale, spread over ten classes: T-shirt, Trouser, Pullover, Dress, Coat, Sandal, Shirt, Sneaker, Bag and Ankle boot [7]. For the AE and VAE the pixel values were scaled to [0, 1] with ToTensor. For the DCGAN they were normalised to [−1, 1] to match the Tanh output of the generator. All three trained models used a batch size of 128. No random seed was set, so a re-run will give slightly different numbers.

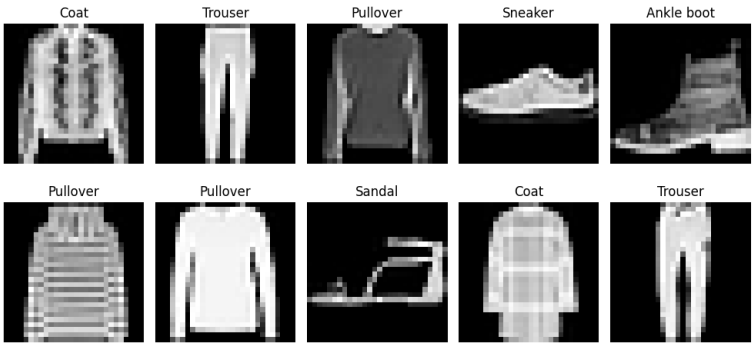


Fig. 1 Sample Fashion-MNIST training images (dataset preview, with class labels).

4.2 Model architectures

Model	Layers (as printed by the notebook)	Output	Parameters
Autoencoder	Encoder: Flatten, Linear 784→256, ReLU, Linear 256→64, ReLU. Decoder: Linear 64→256, ReLU, Linear 256→784, Sigmoid, Unflatten.	28 × 28 image	435,536
VAE	Linear 784→400 + ReLU; two heads Linear 400→20 (mean, log-variance); reparameterise; Linear 20→400 + ReLU; Linear 400→784 + Sigmoid.	28 × 28 image	652,824
DCGAN generator	Linear 100→6272, BatchNorm1d, ReLU, reshape to 128×7×7; ConvTranspose2d 128→64 (4×4, stride 2), BatchNorm2d, ReLU; ConvTranspose2d 64→1 (4×4, stride 2), Tanh.	28 × 28 image	778,305
DCGAN discriminator	Conv2d 1→64 (4×4, stride 2), LeakyReLU(0.2); Conv2d 64→128 (4×4, stride 2), BatchNorm2d, LeakyReLU(0.2); Flatten; Linear 6272→1, Sigmoid.	Real/fake probability	138,817
Stable Diffusion v1.5	Pretrained text encoder, U-Net denoiser and autoencoder decoder (runwayml/stable-diffusion-v1-5, loaded in float16).	512 × 512 colour image	Not counted here

Table 2 Architectures used. Parameter counts for the four trained networks were calculated from the printed layer sizes (weights and biases, plus BatchNorm scale and shift).

4.3 Training settings

Setting	AE	VAE	DCGAN	Stable Diffusion
Optimiser	Adam	Adam	Adam (β1 = 0.5, β2 = 0.999) for G and D	Not trained
Learning rate	0.001	0.001	0.0002	n/a
Epochs / batch size	10 / 128	10 / 128	10 / 128	n/a
Loss	MSE	BCE (sum) + KL	BCE (adversarial)	n/a
Latent size	64	20	100 (noise)	Latent diffusion
Inference settings	n/a	n/a	64 samples from noise	20 steps, guidance 7.0, 512 × 512

Table 3 Training and inference settings taken from the notebook.

4.4 Stable Diffusion configuration

The pipeline was loaded with the diffusers library in half precision (float16) with attention slicing to save memory. The safety checker was disabled by passing safety_checker=None; the library warns that the licence conditions still apply, and the images were used only for this study. Four prompts were used, one at a time:

- **Prompt 1:** A realistic red sports sneaker, professional product photography, white background
- **Prompt 2:** A blue denim jacket displayed on a mannequin, fashion photography
- **Prompt 3:** A black leather handbag, luxury product photography, studio lighting
- **Prompt 4:** A pair of modern white running shoes, clean product photography

Scope note: AE, VAE and DCGAN work on 28 × 28 grayscale Fashion-MNIST images, whereas Stable Diffusion generates 512 × 512 colour images from text and was not trained on Fashion-MNIST here. The study therefore compares model capabilities and output types; it is not a controlled, same-domain image-quality benchmark.

5. Results and Analysis

5.1 Training behaviour

Table 4 lists the mean loss of each epoch printed by the notebook. The AE loss is the batch-average MSE per pixel. The VAE loss is the sum of cross-entropy and KL divergence per image. The DCGAN values are the average discriminator and generator losses. These three losses have different definitions and scales and must not be compared with each other.

Epoch	AE loss (MSE)	VAE loss (per image)	DCGAN D loss	DCGAN G loss
1	0.033584	284.9157	0.3026	2.7064
2	0.017143	255.1615	0.2800	2.7879
3	0.014579	249.8574	0.3726	2.4918
4	0.013017	247.3438	0.5012	2.1257
5	0.012182	245.8108	0.5527	1.9791
6	0.011358	244.7123	0.5774	1.9147
7	0.010856	243.9876	0.5720	1.9383
8	0.010364	243.2970	0.5594	1.9854
9	0.010010	242.7599	0.5616	2.0069
10	0.009718	242.3201	0.5507	2.0316

Table 4 Mean training loss per epoch (values printed by the notebook).

The AE loss falls quickly in the first three epochs (from 0.033584 to 0.014579) and then improves slowly to 0.009718, which suggests that more epochs would still help a little. The VAE loss shows the same pattern, dropping from 284.9157 to 242.3201. For the DCGAN, the discriminator loss rises from 0.3026 to about 0.55 while the generator loss falls from 2.7064 to about 2.0 and then stays level from around epoch 5. This means the discriminator is no longer winning easily, which is a healthier balance than at the start, but the losses by themselves do not show how good the images are.

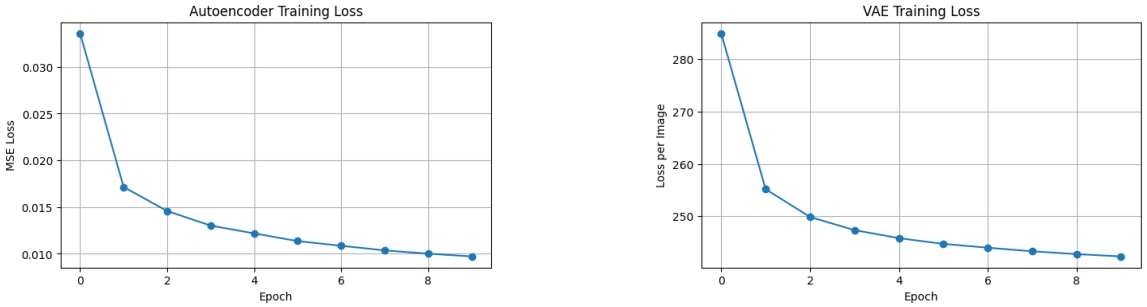


Fig. 2 Training loss curves. Left: AE (MSE). Right: VAE (loss per image). The epoch axis starts at 0, so index 0 is epoch 1.

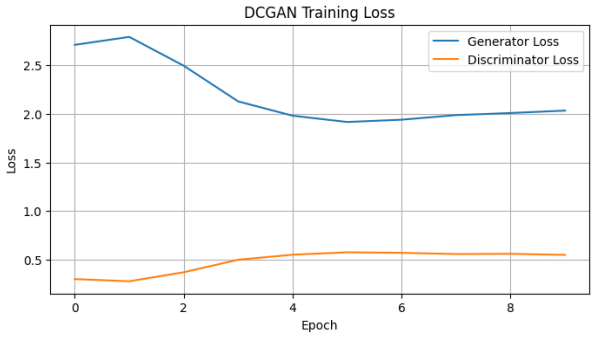


Fig. 3 DCGAN generator and discriminator loss per epoch.

5.2 Reconstruction: AE versus VAE (CO3)

Both models received the same ten test images. Figures 4 and 5 show the originals in the top row and the reconstructions in the bottom row.

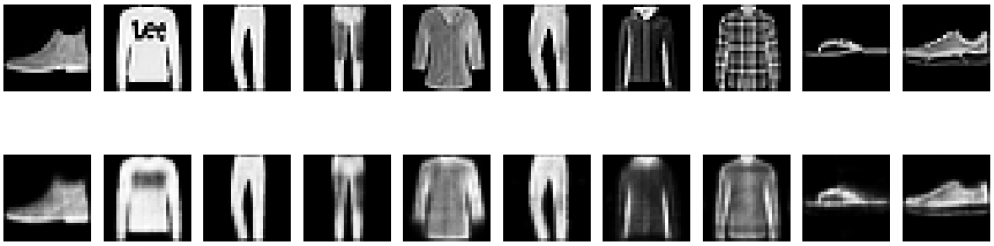


Fig. 4 Autoencoder: original (top) vs reconstructed (bottom), first 10 test images.

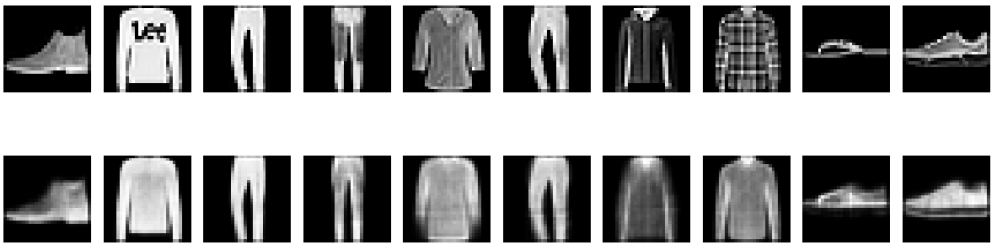


Fig. 5 VAE: original (top) vs reconstructed (bottom), same 10 test images.

Observations for the AE. All ten items keep their overall silhouette: the ankle boot, sweater, trousers, shirt and sneaker are all recognisable. Fine detail is softened. The printed logo on the second image becomes a smeared dark band, the plaid pattern on the eighth image fades to a faint texture, and the thin sandal in the ninth image is blurred with a ghost-like halo.

Observations for the VAE. The silhouettes are also kept, but the outputs are visibly smoother and lower in contrast than the AE outputs. The logo disappears completely, the plaid shirt becomes a plain garment, and the striped shirt in the seventh position turns into a soft blob. The sandal and sneaker in the last two positions keep their basic outline but lose their internal detail. The trousers, which have simple shapes, are reconstructed well by both models.

Model	Test MSE (10,000 images)	Latent size
Autoencoder	0.009628	64
VAE	0.015939	20

Table 5 Reconstruction error on the full Fashion-MNIST test set (notebook cell for test MSE).

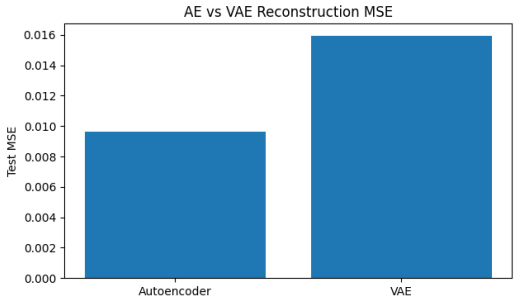


Fig. 6 AE vs VAE reconstruction MSE (bar chart from the notebook).

The AE has the lower test MSE, which agrees with the visual impression that its outputs are sharper. Two factors explain much of the gap. First, the VAE reconstructs from a randomly sampled latent vector, which adds noise. Second, the KL term limits how much information the latent code may carry. The VAE also uses a smaller latent vector (20 values against 64) and a different loss, so the comparison is not a clean test of one idea alone. The result comes from one training run and should be read as an observation, not as a general ranking.

5.3 Generation: DCGAN (CO4)



Fig. 7 DCGAN: 64 generated samples after 10 epochs.

After 10 epochs the generator produces clearly recognisable fashion items: sneakers and boots, long-sleeved tops, T-shirts and trousers. Edges are sharper than in the VAE reconstructions, and the samples are not limited to a single class. Many samples are still poor. Several tops look alike, some images show grainy dark blobs, and a few shapes are broken or hard to name. This is consistent with a small network trained for only 10 epochs. We did not compute a diversity or quality score, so we cannot say whether mode collapse occurred; we only note that several tops look alike.

5.4 Generation: Stable Diffusion (CO4)

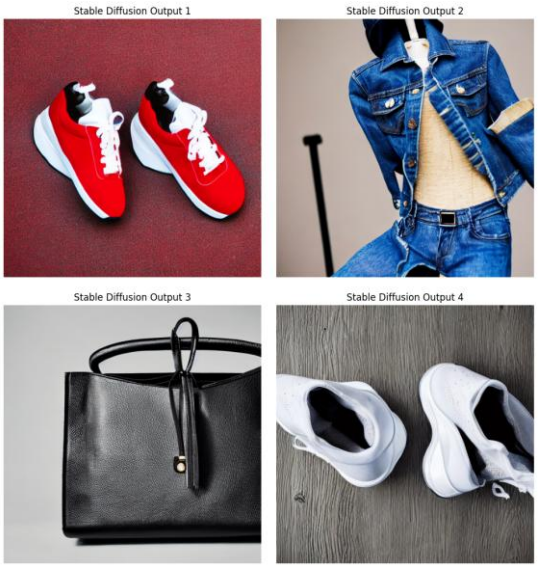


Fig. 8 Stable Diffusion outputs: red sneakers, denim jacket, leather handbag, white running shoes (left to right, top to bottom).

#	Prompt (short)	Observation
1	Red sports sneaker, white background	Photo-like red and white sneakers with clean laces and soft shading. The background is a red textured surface instead of the requested white, and the collar areas look odd.
2	Denim jacket on a mannequin	Clear stitching, buttons and fabric folds. The mannequin looks partly like a torso with bare skin-tone fabric, and a stray black pole appears beside it.
3	Black leather handbag, studio lighting	Convincing leather grain and soft studio light. The handle and strap loop look slightly tangled and the bag is cropped at the edge.
4	White running shoes, clean photography	Realistic mesh texture. The two shoes overlap awkwardly, the laces are unclear and the background is a grey wooden floor rather than clean white.

Table 6 Per-image observations for the four Stable Diffusion outputs.

All four images are colour, high resolution and photo-like, and none of the other models comes close in detail or realism. The weaknesses are of a different kind from the blur seen in AE and VAE: the model follows the general subject of each prompt but misses some details (the white background in prompts 1 and 4), and it can produce small anatomical or structural oddities (the mannequin in image 2, the overlapping shoes in image 4). No numerical quality score was recorded for DCGAN or Stable Diffusion.

6. Comparative Discussion

The four models answer different questions, so the comparison below is qualitative and based on what we observed in this study.

Criterion	AE	VAE	DCGAN	Stable Diffusion
How output is judged	Match to the input image (MSE 0.009628)	Match to the input image (MSE 0.015939) and latent smoothness	Whether the discriminator accepts the sample; visual inspection	Realism and agreement with the prompt
Sharpness seen	Soft	Softest	Fairly sharp edges	Sharp and detailed
Can create new images	Not designed to	Yes, by sampling (not shown in our notebook)	Yes, from noise	Yes, from text
Training effort here	10 epochs, one loss	10 epochs, one loss	10 epochs, two networks and balance needed	None (pretrained)
Control over output	Input image only	Input image or latent values	Almost none (random noise)	Text prompt
Main weakness seen	Blur, lost texture	Stronger blur, lost logos	Noisy and broken samples	Prompt mismatches, odd shapes, GPU need

Table 7 Qualitative comparison based on the outputs of this study.

AE and VAE are judged by how closely the output matches a given input, so reconstruction error is meaningful. A VAE additionally shapes its latent space so that random sampling is possible, but this regularisation tends to give blurrier output than a plain AE, as our test MSE and figures show. Our notebook did not decode random latent vectors for the VAE, so we do not report VAE sampling quality.

DCGAN has no input image to match; the generator is judged only by whether the discriminator accepts its samples. This gives sharper edges than the VAE but less stable training and some broken samples, and the losses do not tell us how good the images are. Stable Diffusion adds text conditioning, so its images are judged by realism and by how well they follow the prompt, which is why no reconstruction MSE applies to it. Its quality comes at the cost of a large pretrained model and a GPU, and the user has less control than the word “prompt” suggests, because details such as the background can be ignored.

Choosing between the models therefore depends on the goal. For compression, denoising or anomaly detection, an AE is simple and cheap. When a smooth latent space and sampling are needed, a VAE is the natural choice. A DCGAN is a light way to create new samples in a small domain, if enough training time is available. When realistic, controllable images are required, a pretrained diffusion model is the strongest option of the four.

7. Limitations and Future Work

7.1 Limitations

- Training lasted only 10 epochs per model and the notebook was run once, so results may change with more training or another run (no seed was set).
- The models work on different data: Stable Diffusion generates colour images from text and was not trained on Fashion-MNIST here, so the study is not a same-domain quality benchmark.

- Only MSE was measured for reconstruction. No numerical quality or diversity score (such as FID) was computed for the DCGAN or Stable Diffusion, so those judgements are visual.
- Only ten test images were displayed for AE and VAE, and four prompts were used for Stable Diffusion.
- The AE and VAE differ in latent size (64 against 20) and in loss, so the MSE gap is not caused by one design choice alone.
- The Stable Diffusion safety checker was disabled for this experiment.

7.2 Future work

- Train for more epochs and with several random seeds, and report the spread of the results.
- Use the same latent size for AE and VAE, and decode random latent vectors from the VAE to study its sampling quality.
- Measure generation quality with a standard metric such as FID, and check diversity of the DCGAN samples.
- Try more prompts and prompt variations for Stable Diffusion, and compare guidance scales and step counts.

8. Conclusion

Reconstruction models (AE, VAE) are simple and suit compression and denoising but blur detail, and the VAE blurred more than the AE in our run (test MSE 0.015939 against 0.009628). The DCGAN can create new clothing images with fairly sharp edges but needs more training for clean results. The pretrained diffusion model produced the most detailed and realistic images and responds to text, at a higher computational cost and with prompt mismatches and minor artefacts. No model is best for every task: the right choice depends on whether the goal is faithful reconstruction, sampling, or controllable generation. The study is limited by 10-epoch training, a single run, a small set of displayed samples, and different data across the models.

9. AI Usage Documentation and Group Reflection

- **AI tools used:** ChatGPT (OpenAI) and Claude (Anthropic) were used to understand the models, read the notebook outputs and draft and format this report.
- **Important prompts (ChatGPT):**

  1. “Provide complete beginner-friendly Google Colab code to implement Autoencoder (AE), Variational Autoencoder (VAE), DCGAN, and Stable Diffusion using the Fashion-MNIST dataset.”

  2. “Explain how to train an Autoencoder and Variational Autoencoder on Fashion-MNIST and display original versus reconstructed images with training loss graphs.”

  3. “Help debug DCGAN image generation, Stable Diffusion model loading, and calculate the reconstruction MSE for AE and VAE. Provide corrected Google Colab code.”
- **Verification:** losses, test MSE values, model sizes and settings were checked against the notebook’s printed output; all figures are the notebook’s own output images; observations were written after viewing each figure; model descriptions follow the references below.
- **Key learning:** reconstruction and generation are different goals; a reconstruction error alone cannot rank unlike models; adversarial losses do not measure image quality; pretrained models allow high-quality generation without training from scratch.

Contribution of each group member

Group member	Contribution (2–3 lines each)
Nilesh Chavan (202401110017)	Collected and explored the Fashion-MNIST dataset and implemented the Autoencoder and Variational Autoencoder models. Trained the models, evaluated image reconstruction using MSE, and captured the output screenshots.

Group member	Contribution (2–3 lines each)
Bhusan Bhamre (202401110078)	Implemented and trained the DCGAN model and generated new fashion images. Used the pretrained Stable Diffusion model to generate fashion images, documented the results, and helped prepare the final report and presentation.

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